Short Summary: Semi-Supervised Autoencoder (Lion vs. Tiger)

Objective: Train an autoencoder on CIFAR-100 for unsupervised feature learning and use these features to classify lions vs. tigers with limited labelled data (semi-supervised).

Method:

1. **Autoencoder Training:** Trained a convolutional autoencoder on all CIFAR-100 images for **15 epochs** with a **128-dimensional latent space**.
2. **Feature Extraction:** Used the trained encoder to generate feature vectors for lion and tiger images.
3. **Classification:** Trained a logistic regression classifier using features from **50 labeled examples** per class (lion and tiger).
4. **Evaluation:** Tested the classifier on the CIFAR-100 test set images for lions and tigers.

Results:

- Final Autoencoder Loss: **0.0492** (decreased steadily from ~0.1035).
- Classifier Accuracy: **56.5%**.
- F1 Scores: Lion = 0.58, Tiger = 0.54.
- AUC: 0.598.
- Reconstructions showed basic shape/color learning. t-SNE showed feature overlap.

Discussion: The decreasing loss indicates that the autoencoder learnt meaningful representations. With just 100 labelled examples, the classifier's performance was marginally better than chance, suggesting that the learnt features were somewhat helpful but constrained by the short training period, small latent size, and small labelled set. There was a noticeable overlap in features between classes. Increasing the number of labelled samples for the classifier, the size of the latent dimension, and the AE training epochs could probably boost performance.